ATOMIC-LEVEL MATERIALS SIMULATION

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Kenan Song
School of Chips
Email:
Kenan.Song@xjtlu.edu.cn
Office: C5007

Non-Equilibrium Green Function Foundations

I. Open Boundary Partitioning and The Emergence of Self Energy (Σ^R)

II. The Fisher-Lee Relation and Coherent Resonant Transmission

III. Equilibrium v.s. Non-Equilibrium: Kinetic Population and Keldysh Equation

I. Open Boundary Partitioning and The Emergence of Self Energy (Σ^R)

♦ **The Landauer formula requires the transmission spectrum $T(E)$, but lacks an explicit recipe to calculate it for complex, atomistic, or non-equilibrium channels**

♦ **The Open Boundary Problem: A real transistor or molecular junction is an open quantum system connected to infinite leads. The Hamiltonian matrix is infinite-dimensional ($\infty\times\infty$)**

♦ **Use the Green's function formalism to project an infinite open system down into an exact, finite-dimensional matrix problem focused exclusively on the central device region**

It was established that the current can be obtained once the transmission probability $T(E)$ is given. The remaining task is to determine how $T(E)$ is actually calculated for a device containing thousands of atoms under a strong bias voltage. Standard quantum mechanics textbooks describe the solution of the Schrödinger equation for closed systems (such as a particle in a box). A device, however, is an open system, with electrons constantly entering and leaving. Consequently, the Hamiltonian matrix is infinite. The Non-Equilibrium Green's Function (NEGF) formalism is introduced here as the exact mathematical tool to map this infinite open system onto a manageable, finite matrix

I. Open Boundary Partitioning and The Emergence of Self Energy (Σ^R)

The Microscopic Propagator: Green's Function

Definition of the differential equation operator for a steady-state system at energy E: $(\boldsymbol{E - H})\psi = \mathbf{0}$

Introduction of the Green's function G as the impulse response (inverse operator) to a spatial Delta function source $\boldsymbol{\delta(r - r')}$**:** $(\boldsymbol{E - H})\boldsymbol{\delta(r, r', E)} = \boldsymbol{\delta(r - r')}$

Matrix representation for a discrete lattice (Tight Binding basis): $(\boldsymbol{EI - H})\boldsymbol{G} = \boldsymbol{I} \rightarrow \boldsymbol{G} = (\boldsymbol{EI - H})^{-1}$

Green's function is the inverse of a differential operator. In quantum mechanics, it serves as a 'microscopic propagator'. For a source at position r/r' that injects an electron into the system at energy E, the Green's function $G(r, r', E)$ yields the probability amplitude of detecting that electron at another position r after propagation through the system's potential landscape. In a discrete matrix representation, such as the Tight Binding model, the determination of the Green's function reduces to a matrix inversion problem

I. Open Boundary Partitioning and The Emergence of Self Energy (Σ^R)

Singularities on the Real Energy Axis

♦ **If E matches an exact eigenvalue (eigenenergy ε_n) of the isolated Hamiltonian H, the matrix $(EI - H)$ becomes singular (determinant equals zero)**

♦ **The inverse matrix $(EI - H)^{-1}$ diverges to infinity at these bound states**

♦ **Solution: Displace the energy E slightly off the real axis into the complex plane by adding an infinitesimal imaginary part, $i\eta$ or $-i\eta$ (where, $\eta \to 0^+$)**

A critical mathematical hurdle arises in the inversion of the matrix $(EI - H)$. When the energy E coincides with an intrinsic energy level of the system, the matrix becomes singular — its determinant vanishes, preventing inversion. To circumvent this, the energy is shifted slightly off the real axis into the complex plane via the addition of a tiny imaginary term ($i\eta$ or $-i\eta$). This procedure splits the Green's function into two distinct, physically meaningful variants

I. Open Boundary Partitioning and The Emergence of Self Energy (Σ^R)

Causal Propagation: Retarded (G^R) and Advanced (G^A) Formulations

Definition of the Retarded Green's Function (G^R): $G^R = \lim_{\eta \to 0^+} [(E + i\eta)I - H]^{-1}$

Definition of the Advanced Green's Function (G^A): $G^A = \lim_{\eta \to 0^+} [(E - i\eta)I - H]^{-1}$

Fundamental algebraic relationship between them: $G^R = \left(G^A\right)^\dagger$

A shift of the energy into the upper half of the complex plane ($i\eta$) defines the Retarded Green's Function (G^R). In the time domain, G^R represents forward-in-time causality, describing the evolution after an electron is injected into the system. Conversely, a shift into the lower half ($-i\eta$) yields the Advanced Green's Function (G^A), which tracks reverse causality. The Hermiticity of the Hamiltonian matrix renders G^R and G^A conjugate transposes of each other

I. Open Boundary Partitioning and The Emergence of Self Energy (Σ^R)

The Local Density of States (LDOS)

The spectral function operator $A(E)$ is defined as: $A(E) = i\left[G^R(E) - G^A(E)\right]$

Using the algebraic identity $\lim_{\eta\to 0^+} \frac{1}{x \pm i\eta} = \mathcal{P}\left(\frac{1}{x}\right) \mp i\pi\delta(x) \rightarrow \text{Trace}[A(E)] = 2\pi \sum_n \delta(E - \varepsilon_n) = 2\pi D(E)$

The diagonal elements of $A(E)$ give the Local Density of States (LDOS) at a specific atom/orbital m: $\rho_m(E) = \frac{1}{2\pi} A_{mm}(E)$

The Retarded Green's function contains the complete information of the system's quantum states. Subtracting G^A from G^R and multiplying by i yields a highly critical operator called the Spectral Function, $A(E)$. The diagonal elements of this matrix map out the Local Density of States (LDOS), providing exact information on how the available electronic states at energy E are distributed across different atoms in space

I. Open Boundary Partitioning and The Emergence of Self Energy (Σ^R)

Partitioning the Total System Block Matrix



$$H_{total} = \begin{pmatrix} H_L & H_{LC} & 0 \\ H_{CL} & H_C & H_{CR} \\ 0 & H_{RC} & H_R \end{pmatrix}$$

$H_{L/R}$ — Infinite matrices representing the semi-infinite periodic leads

H_C — Finite matrix representing the central device region

$H_{LC/RC}$ — Coupling matrices linking the left/right leads to the device

$H_{LC} = (H_{CL})^\dagger$; $H_{CR} = (H_{RC})^\dagger$

Assume no direct coupling between Left and Right leads: $H_{LR} = 0$

The simulation of an open quantum device begins by organizing the infinite Hamiltonian into a structured block matrix. Space is partitioned into three regions: the infinite Left Lead (H_L), the finite Central Device (H_C), and the infinite Right Lead (H_R). Electron hopping between the leads and the device occurs via the coupling matrices (H_{LC} and H_{CR}). Crucially, the leads are assumed to be sufficiently separated to preclude any direct quantum hopping between the left and right contacts without passing through the device

I. Open Boundary Partitioning and The Emergence of Self Energy (Σ^R)

Matrix Algebra of the Open Boundary Problem

Write out the full equation for the Retarded Green's function using the block structure

$$\begin{pmatrix} (E+i\eta)I - H_L & -H_{LC} & 0 \\ -H_{CL} & (E+i\eta)I - H_C & -H_{CR} \\ 0 & -H_{RC} & (E+i\eta)I - H_R \end{pmatrix} = \begin{pmatrix} H_L & H_{LC} & H_{LR} \\ H_{CL} & H_C & H_{CR} \\ H_{RL} & H_{RC} & H_R \end{pmatrix} = \begin{pmatrix} I & 0 & 0 \\ 0 & I & 0 \\ 0 & 0 & I \end{pmatrix}$$

Isolate the middle row to solve explicitly for G_C, the Green's function within the finite device domain

Performing the matrix multiplication for the center column of the identity system yields the extraction of G_C, the Green's function confined to the finite central device region. Through the setup of simultaneous matrix equations across the partitioned rows, the infinite lead variables are systematically isolated and eliminated

I. Open Boundary Partitioning and The Emergence of Self Energy (Σ^R)

Eliminating Infinite Leads: The Mathematical Emergence of Self-Energy

From the system, extract the two flanking relationships:

$$-\boldsymbol{H_{CL}G_{LC}} + [(\boldsymbol{E}+\boldsymbol{i\eta})\boldsymbol{I} - \boldsymbol{H_C}]\boldsymbol{G_C} - \boldsymbol{H_{CR}G_{RC}} = \boldsymbol{I}$$

$$[(\boldsymbol{E}+\boldsymbol{i\eta})\boldsymbol{I} - \boldsymbol{H_L}]\boldsymbol{G_{LC}} - \boldsymbol{H_{LC}G_C} = \boldsymbol{0} \Rightarrow \boldsymbol{G_{LC}} = [(\boldsymbol{E}+\boldsymbol{i\eta})\boldsymbol{I} - \boldsymbol{H_L}]^{-1}\boldsymbol{H_{LC}G_C}$$

Substitute these back into the central equation to isolate G_C:

$$\{(\boldsymbol{E}+\boldsymbol{i\eta})\boldsymbol{I} - \boldsymbol{H_C} - \boldsymbol{H_{CL}}[(\boldsymbol{E}+\boldsymbol{i\eta})\boldsymbol{I} - \boldsymbol{H_L}]^{-1}\boldsymbol{H_{LC}} - \boldsymbol{H_{CR}}[(\boldsymbol{E}+\boldsymbol{i\eta})\boldsymbol{I} - \boldsymbol{H_R}]^{-1}\boldsymbol{H_{RC}}\}\boldsymbol{G_C} = \boldsymbol{I}$$

Define the Retarded Self-Energy for contact $\alpha \in \{L, R\}$: $\Sigma_\alpha^R(E) = \boldsymbol{H_{C\alpha}}[(\boldsymbol{E}+\boldsymbol{i\eta})\boldsymbol{I} - \boldsymbol{H_\alpha}]^{-1}\boldsymbol{H_{\alpha C}}$

Substitution of the lead expressions back into the central block equation yields a remarkable result. The infinite dimensions of the leads disappear, leaving behind a highly elegant corrections term attached to the finite device Hamiltonian. This term is the Self Energy (Σ^R). The self energy encapsulates the entire effect of the infinite lead on the finite device, serving as an effective potential that accounts for the open boundaries

I. Open Boundary Partitioning and The Emergence of Self Energy (Σ^R)

Self Energy: Shifts and Broadening

The effective Retarded Green's function for the finite central device is now: $G_c^R(E) = \left[E \cdot I - H_C - \Sigma_L^R(E) - \Sigma_R^R(E)\right]^{-1}$

Deconstruct Σ^R **into its real and imaginary parts:** $\Sigma^{\pi}(E) = \Delta E - i\frac{\Gamma(E)}{2}$

The Real part of $\Delta(E)$**: Induces an energy shift in the device's molecular levels due to the chemical connection with the lead**

The Imaginary part of $\Gamma(E)$**: Induces life-time broadening of the energy states**

Look closely at our new formula for G_c^R**. The infinite leads have been completely replaced by two finite matrices:** Σ_L^R **and** Σ_R^R**. Because these matrices are complex value, they do two things. The real part,** $\Delta(E)$**, acts as a potential shift that modifies the internal energy levels of the device. The imaginary part,** $\Gamma(E)$**, acts as an energy broadening term. This broadening reflects the finite lifetime of an electron inside the device before it escapes back into the leads**

I. Open Boundary Partitioning and The Emergence of Self Energy (Σ^R)

The Broadening Matrix $\Gamma(E)$ and Particle Escape Rates:

Formal definition of the Level-Width Matrix Γ_α for lead α: $\Gamma_\alpha(E) = i\left[\Sigma_\alpha^R(E) - \Sigma_\alpha^A(E)\right]$

Since $\Sigma^R = \left(\Sigma^A\right)^\dagger$, this can be written as: $\Gamma_\alpha(E) = -2 \cdot Imag[\Sigma_\alpha^R(E)]$

Physical Interpretation: Connection to the quantum mechanical lifetime (τ) of an electron state via the uncertainty principle: $\Gamma \approx \hbar/\tau$

Formalization of the imaginary part of the self-energy proceeds through the definition of the Broadening Matrix (Γ_α). Being directly proportional to the imaginary component of the self energy, Γ maps out how strongly an electronic state couples to the outside world. According to the Heisenberg uncertainty principle, a larger Γ corresponds to a shorter lifetime (τ) for an electron residing inside the channel before leaking out into the lead

II. The Fisher-Lee Relation and Coherent Resonant Transmission

The Microscopic Formula for Transmission $T(E)$:

The Fisher-Lee Formula for the transmission coefficient: $$T(E) = Trace\left[\Gamma_L(E) G_C^R(E) \Gamma_R(E) G_C^A(E)\right]$$

Dimensional match verification: Γ and G are matrices of size ($N \times N$), where N is the number of localized basis functions in the central device. The Trace collapses this matrix product down to a single dimensionless scalar value

The derivation arrives at one of the absolute pillars of modern transport theory: the Fisher-Lee Formula. This equation bridges the gap between last two lectures, proving that the abstract transmission probability $T(E)$ from the Landauer formula can be calculated explicitly by multiplying the broadening matrices (Γ) of the contacts with the Retarded (G^R) and Advanced (G^A) Green's functions of the device, and taking the Trace of the resulting matrix

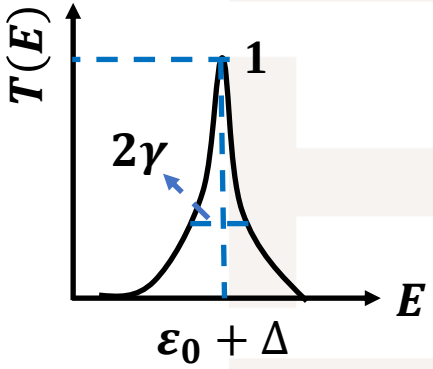
II. The Fisher-Lee Relation and Coherent Resonant Transmission

Graphical Analysis of: $T(E) = Trace\left[\Gamma_L(E) G_C^R(E) \Gamma_R(E) G_C^A(E)\right]$

Example: A single-level quantum dot system with site energy ε_0 coupled equally to two leads ($\Gamma_L = \Gamma_R = \gamma$)

Analytical reduction of the Fisher-Lee matrix equation into a scalar Breit-Wigner formula:

$$T(E) = \frac{\gamma^2}{(E - \varepsilon_0 - \Delta)^2 + \gamma^2}$$



Draw the Lorentzian transmission peak on the blackboard, highlighting the peak value of 1 at resonance ($E = \varepsilon_0 + \Delta$) and a Full Width at Half Maximum (FWHM) equal to 2γ

The physical insight behind the Fisher-Lee formula can be understood through a simple example: a single atom sandwiched between two contacts. Reduction of the matrix formula to this basic system simplifies it into a classic Lorentzian resonance curve. This demonstrates that when an incoming electron's energy matches the broadened energy level of the atom, the transmission probability reaches exactly 1 — a phenomenon known as resonant tunneling

III. Equilibrium v.s. Non-Equilibrium: Kinetic Population and Keldysh Equation

Non-Equilibrium case: G^R and G^A are not enough

♦ G^R and G^A describe the available states and energy spectrum, but they do not contain information about how electrons fill those states

♦ In equilibrium, state occupancy is determined by the Fermi-Dirac distribution

♦ Under a large external voltage bias, the system is driven out of equilibrium, meaning the state occupancy can no longer be described by a single global Fermi level

♦ Introduction of the Lesser Green's Function ($G^<$) to track actual electron density

All quantities derived so far (G^R, G^A, Γ and $T(E)$) describe the structural backbone of the device — namely, the available channels and quantum states. However, knowledge of available states is insufficient; it is also necessary to determine whether those states are occupied or empty. In equilibrium, the Fermi-Dirac distribution suffices. Under a strong voltage bias, however, the device is driven far out of equilibrium. To track the dynamical population of these states, a new Green's function is required: the Lesser Green's Function ($G^<$)

III. Equilibrium v.s. Non-Equilibrium: Kinetic Population and Keldysh Equation

The Density Matrix Operator:

Formal definition of the Lesser Green's Function matrix in the energy domain: $G^{<}(E) \equiv i\langle c_j^{+} c_i \rangle_E$

The fundamental relation between $G^{<}(E)$ **and the real space steady-state Electron Density Matrix (**ρ**):**

$$\rho = \frac{1}{2\pi}\int_{-\infty}^{\infty} Imag[G^{<}(E)]dE$$

The diagonal elements ρ_{ii} **give the net electron population residing on atom** i **under arbitrary non-equilibrium bias conditions**

The Lesser Green's Function ($G^{<}$**) tracks electron statistics, with its underlying definition built from quantum creation and annihilation operators. Integration of** $G^{<}$ **over the entire energy axis yields the steady-state Electron Density Matrix (**ρ**). The diagonal values of this matrix provide the exact occupation numbers on each atom under non-equilibrium conditions**

III. Equilibrium v.s. Non-Equilibrium: Kinetic Population and Keldysh Equation

The Keldysh Equation: Solving for $G^<$

Presentation of the Keldysh Equation: $G^<(E) = G^R(E)\Sigma^<(E)G^A(E)$

Introduction of the Lesser Self Energy ($\Sigma^<(E)$), which acts as the electron injection source from the contacts

Formula for the total lesser self-energy from all contacts: $\Sigma^<(E) = \Sigma_L^<(E) + \Sigma_R^<(E) = i\Gamma_L(E)f_L(E) + i\Gamma_R(E)f_R(E)$

The calculation of $G^<$ proceeds through the Keldysh Equation. This equation states that the actual electron population inside the device ($G^<$) is determined by propagating the electron streams injected from the leads ($\Sigma^<$) through the device channel using the structural matrices (G^R and G^A). The lesser self energy $\Sigma^<$ is given by the coupling matrix Γ scaled by the Fermi-Dirac occupancy factor of the respective lead

III. Equilibrium v.s. Non-Equilibrium: Kinetic Population and Keldysh Equation

Decomposing $G^<$ into Localized Source Streams

Substitute the component terms of $\Sigma^<$ into the Keldysh Equation: $G^<(E) = G^R[i\Gamma_L f_L + i\Gamma_R f_R]G^A$

Separate the expression into its left-injected and right-injected contributions: $G^<(E) = iG^R\Gamma_L G^A f_L(E) + iG^R\Gamma_R G^A f_R(E)$

Physical intuition: The local electron population is a weighted superposition of states filled by the Left lead and states filled by the Right lead

The expression separates cleanly into two components. The first component represents electrons injected from the Left Lead, moving with an occupancy determined by f_L. The second component represents electrons injected from the Right Lead, moving with an occupancy determined by f_R. The total electron population at any point within the non-equilibrium device is given by a simple, weighted superposition of these two distinct inbound carrier streams

III. Equilibrium v.s. Non-Equilibrium: Kinetic Population and Keldysh Equation

Deriving the Microscopic Current Operator via Particle Conservation:

The net current I_α entering the central device from lead α must equal the net rate of change of the electron population injected from that contact

Using the quantum mechanical operator equation of motion for total charge: $I_\alpha = e\left\langle \frac{dN_\alpha}{dt} \right\rangle = \frac{ie}{\hbar}\langle [H, N_\alpha] \rangle$

Translate this expectation value into Green's function matrices:

$$I_\alpha = \frac{e}{h}\int_{-\infty}^{\infty} Trace\left[\Sigma_\alpha^<(E)G^R(E) - \Sigma_\alpha^A(E)G^<(E) + h.c.\right]dE$$

The exact electrical current under non-equilibrium conditions is obtained by invoking particle conservation. The current passing through a contact interface must equal the net rate of change of the electron population injected from that contact. By expressing the quantum mechanical commutator for the particle number operator, an exact matrix equation for the current is derived, formulated directly in terms of the Green's functions and self-energies

III. Equilibrium v.s. Non-Equilibrium: Kinetic Population and Keldysh Equation

The Mathematical Equivalence of NEGF and Landauer Current:

Simplify the general NEGF current equation for Lead L: $I_L = \frac{e}{h}\int_{-\infty}^{\infty} Trace\left[\Sigma_L^<(E)G^R(E) - \Sigma_L^A(E)G^<(E) + h.c.\right]dE$

Substitute the Keldysh relation $G^< = G^R\Sigma^<G^A$ **and the identities for** Γ **into this equation**

Through systematic matrix algebra (to be completed on the board), reduce the trace down to:

$$Trace\left[\Sigma_L^<(E)G^R(E) - \Sigma_L^A(E)G^<(E) + h.c.\right] = Trace\left[\Gamma_L(E)G^R(E)\Gamma_R G^A(E)\right] \times [f_L(E) - f_R(E)]$$

Finally: $I_L = \frac{2e}{h}\int_{-\infty}^{\infty} Trace\left[\Gamma_L(E)G^R(E)\Gamma_R G^A(E)\right] \times [f_L(E) - f_R(E)]dE$

Performing the algebraic steps to simplify the current expression, and substituting the expressions for $G^<$ and the self energies, the long matrix equation collapses into a familiar form. The trace simplifies to exactly $Trace\left[\Gamma_L G^R \Gamma_R G^A\right] \times [f_L - f_R]$. This demonstrates that the general NEGF current operator reduces exactly to the Landauer-Büttiker formula, thereby providing an exact analytical bridge between the two formalisms

III. Equilibrium v.s. Non-Equilibrium: Kinetic Population and Keldysh Equation

The Advanced Identity: $G^R - G^A = -iG^R\Gamma G^A$

The Optical Theorem of Quantum Transport:

$$\left(G^R\right)^{-1} - \left(G^A\right)^{-1} = \left(E - H_C - \Sigma^R\right) - \left(E - H_C - \Sigma^A\right) = \Sigma^A - \Sigma^R = i\Gamma$$

Pre-multiply by G^R and post-multiply by G^A to complete the derivation

$$G^R - G^A = -iG^R\Gamma_{total}G^A \quad \text{where, } \Gamma_{total} = \Gamma_L + \Gamma_R$$

Physical meaning: The total state density (left side) must equal the sum of the states injected from the left and right leads combined (right side)

A core matrix identity is proved here to ensure physical validity of the calculations. This identity relates the difference between G^R and G^A to a product involving the total broadening matrix. Physically, it serves as the quantum transport equivalent of the Optical Theorem, guaranteeing that the total available density of states inside the device exactly equals the sum of states injected from the left and right contacts combined, with no leakage

III. Generalization to Multi-Terminal Networks and Topological Applications

Boundary Condition: Coherent v.s. Incoherent Transport

♦ **The derivations presented in this hour assume coherent transport, meaning electrons do not scatter off phonons or experience inelastic electron-electron interactions inside the device**

♦ **Under these conditions, an electron preserves its energy E from the moment it enters the device until the moment it leaves**

♦ **Introducing the concept of Scattering Self Energies ($\Sigma_{scattering}$) as the method to handle inelastic scattering (*e.g.*, electron-phonon coupling) in advanced NEGF simulations**

The boundary conditions of the current framework should be explicitly stated. The equations derived in this hour assume coherent transport, in which electrons are treated as pure waves that traverse the device without energy loss to lattice vibrations (phonons). For devices where heating or phonon scattering is significant, NEGF can accommodate this by the addition of a 'Scattering Self Energy' ($\Sigma_{scattering}$) directly into the matrix loop



That's all for this session

Thank you for your attention and great participation